

NSI-189

Evidence review: mechanisms, pathways, preclinical findings, and qualitative human evidence

BOTTOM LINE

NSI-189 (also called bezisterim) is a research compound with repeated animal signals for neural progenitor proliferation, synaptic plasticity, mitochondrial function, and function in injury or disease models. Its direct molecular target is not established. In people, one 220-person Phase 2 depression trial missed its primary endpoint, despite smaller secondary self-report and selected cognitive signals at 40 mg/day. There is no formal qualitative literature on healthy-person or nootropic use.

Is NSI-189 a credible nootropic?

Verdict: insufficient evidence for nootropic use in healthy people. The compound has a coherent, but target-agnostic, preclinical phenotype. That is distinct from proving it enhances cognition, mood, or resilience in healthy humans. The strongest human dataset is a sponsor-associated 12-week MDD study: its clinician-rated primary endpoint, MADRS, was negative at both 40 and 80 mg/day. Positive self-report outcomes are useful signals, not a confirmation.

Confidence: Weak for a short-term, 40 mg/day self-report signal in selected adults with MDD; **Insufficient** for healthy cognition, subjective nootropic effects, dependence, sexual effects, suicidality, long-term safety, or a usable pharmacovigilance profile.

What NSI-189 is and is not

NSI-189 phosphate is a benzylpiperazine-aminopyridine small molecule developed as a neurogenic, monoamine-reuptake-independent candidate for major depressive disorder. “Neurogenic” is a phenotype claim: experiments report increased neural progenitor proliferation and downstream neuroplasticity markers. It does **not** mean the compound has a demonstrated human hippocampal-growth effect, a defined receptor target, or an established clinical use.

The studies below use a salt form, disease-model animals, and tightly controlled research formulations. They do not validate products sold as research chemicals, whose identity, purity, and dose are separate unknowns.

Mechanism and pathway map

Step	Evidence actually shown	How to interpret it
1. Primary target	No validated receptor, enzyme, or transporter target was identified in the primary NSI-189 papers reviewed. It is described as independent of monoamine reuptake pathways.	The mechanism remains target-agnostic. Do not turn downstream changes into a claim that it is a TrkB agonist, AMPK agonist, or BDNF drug.
2. Neural-progenitor / structural phenotype	Human hippocampus-derived neural stem cells and mouse work are cited as showing pro-proliferative / pro-neurogenic activity. In mice, NSI-189 was pro-proliferative but P7C3-A20 produced a larger net neurogenesis effect by protecting young neurons from death.	Proliferation and survival are different mechanisms. The direct cell-type, dose, and exposure bridge to humans is not established.
3. Injury / trophic signals	In oxygen-glucose-deprivation / reperfusion hippocampal cells, NSI-189 reduced cell death, increased Ki67 and MAP2, and increased BDNF and stem-cell factor (SCF) expression. In Angelman-model hippocampal slices / mice, effects were associated with TrkB and Akt activation.	BDNF-SCF and TrkB-Akt are measured associates in specific models, not proven initiating targets or universal pathways.
4. Synapses and circuit function	Acute mouse hippocampal slices showed time- and dose-dependent enhancement of theta-burst LTP. Diabetic animal work reported synaptic markers, hippocampal neurogenesis, and memory protection.	This is the best mechanistic bridge to cognition, but is still animal / ex-vivo evidence.

Step	Evidence actually shown	How to interpret it
5. Mitochondria / peripheral nerve	In diabetic rats, 30 mg/kg orally for 16 weeks restored DRG AMPK phosphorylation and increased cortical respiratory-complex I / IV activity plus complex III / V subunits; it coincided with improved neuropathy and memory measures.	Associative tissue biology. It does not establish a direct mitochondrial target, and it did not correct hyperglycemia.

Mechanistic chain with confidence: NSI-189 → unknown initiating target → neural-progenitor / trophic and plasticity-associated changes → improved model-specific behavior. The first arrow is unknown; the middle arrows are demonstrated only in selected models; the final behavioral effects are real within those models but are not a human efficacy result.

Preclinical evidence

Study	Model and exposure	Qualitative finding	Main limits
Tajiri et al., 2017 PMID 28181668	Adult Sprague-Dawley rats, MCAO stroke. Oral NSI-189 started 6 h after occlusion and continued 12 weeks.	Motor / neurological deficits improved through 24 weeks. Hippocampal MAP2 neurite marker increased. In OGD/R hippocampal cells: less cell death, more Ki67 / MAP2, higher BDNF and SCF.	Stroke model, not depression or healthy cognition. The abstract does not establish a direct target or human-equivalent exposure.
Parihar et al., 2018 PMID 29351056	Long-Evans rats after fractionated cranial irradiation. Oral daily dosing for 4 weeks.	Better performance on four of five exploration-memory tasks; benefit persisted one month after dosing stopped. Increased neurogenesis and fewer activated microglia; no major hippocampal-volume change.	Radiation injury is a highly specific insult. Multiple behavioral tests create a selective-reporting risk despite internally coherent biology.
Liu et al., 2019 PMID 30408487	Angelman syndrome mouse model; acute hippocampal slices and short daily injections.	Time- and dose-dependent LTP enhancement; motor and cognitive deficits improved. Effects were associated with TrkB and Akt activation.	Genetic neurodevelopmental model. Association with TrkB/Akt does not identify a direct pharmacological target.
Jolivalt et al., 2019 PMID 31492662	Type 1 and type 2 diabetic mice; oral delivery. Also adult rat sensory neurons in culture.	Improved neurite outgrowth and mitochondrial measures in sensory neurons; prevented or halted multiple central and peripheral neuropathy indices, increased hippocampal neurogenesis / synaptic markers / volume, and protected memory.	Disease-specific model; much of the efficacy literature includes a Neuralstem author / supplied compound. Animal findings await independent replication and human testing.
Jolivalt et al., 2022 PMID 35967127	Male Zucker diabetic fatty rats, 16 weeks established diabetes then 30 mg/kg/day orally for 16 weeks; n=10/group for most endpoints.	Restored sensory nerve conduction and other neuropathy measures; improved object-recognition performance. Restored DRG AMPK activity and selective cortical respiratory-complex markers. No glucose or insulin correction.	Strongest mitochondrial detail, but this is long-term disease-model exposure with no animal-to-human PK bridge.

Preclinical “uses” are best read as **research applications**: stroke recovery, radiation-associated cognitive injury, Angelman syndrome, and diabetic central/peripheral neuropathy. They are not validated indications. Positive model results across very different conditions support a broad plasticity phenotype, but they also make target and disease specificity harder to infer.

Human bridge: pharmacokinetics, outcomes, and limits

Study	Design / dose	Key quantitative result	Interpretation
Fava et al., 2016 Phase 1b PMID 26643541	24 adults with recurrent MDD, 28 days inpatient. Three cohorts: 40 mg once, twice, or three times daily; 18 active / 6 placebo. Follow-up to day 84.	Half-life 17.4-20.5 h; steady state 96-120 h. SDQ and CPFQ favoured active treatment; reported effect sizes 0.90 and 0.94. MADRS effect size 0.95 but not statistically significant. Hippocampal-volume differences were not significant.	An encouraging but extremely small study. The MRI result is explicitly descriptive, not human proof of neurogenesis.
Papakostas et al., 2020 Phase 2 SPCD PMID 30626911	220 adults with MDD, 12 weeks. 40 mg/day (n=44), 80 mg/day (n=44), placebo Stage 1 (n=132); placebo nonresponders re-randomized in Stage 2.	Primary MADRS: 40 mg mean difference -1.8 (95% CI -4.8 to 1.1), p=.224; 80 mg -1.4 (-4.4 to 1.5), p=.344. At 40 mg, pooled SDQ -8.2 (-16.2 to -0.2), p=.044; CPFQ -1.9 (-3.7 to -0.1), p=.035. CogScreen showed some positives; Cogstate did not.	The prespecified primary endpoint was null. Secondary self-report and selected cognitive signals justify further research, not a clinical or nootropic conclusion. 80 mg did not show the same pattern.

Concentration realism: the human trial establishes plasma PK but the primary preclinical papers do not supply a compatible animal brain / plasma exposure threshold for a direct concentration comparison. A standard body-surface-area conversion of the 30 mg/kg rat dose is about 4.8 mg/kg human-equivalent, or roughly 336 mg/day for a 70 kg adult. That arithmetic is a toxicology heuristic, **not** a dose recommendation, and does not validate cross-species exposure or benefit.

Qualitative findings and subjective effects

WHAT THE LIVED-EXPERIENCE LITERATURE DOES NOT CONTAIN

No peer-reviewed NSI-189 qualitative interview / focus-group study, case report or case series, patient-community-mining study, or usable published pharmacovigilance signal was identified. An exact openFDA FAERS search returned no NSI-189 reports, which is expected to be uninformative for an investigational, non-marketed exposure with no denominator.

The only structured subjective data come from self-report scales in the two MDD trials. In Phase 2, 40 mg/day improved the SDQ and CPFQ, with item-level signals involving low affect, mood responsiveness, tearfulness, anxiety, decision-making, ability to work, function, optimism, and outlook. Those are patient-reported symptom scores, not interviews or narratives, and they sit beside a null clinician-rated primary endpoint.

Source	Subjective / adverse-event signal	What can responsibly be said
Phase 1b (n=18 active; 6 placebo)	No serious adverse events. Investigator-attributed active vs placebo: headache 50% vs 50%; dizziness 27.8% vs 16.7%; somnolence 27.8% vs 16.7%; dry mouth 11.1% vs 0%; vivid dreams / nightmares 16.7% vs 0%; insomnia 16.7% vs 0%; nausea 11.1% vs 0%.	Small cohort counts are not incidence estimates. At 120 mg/day, dizziness occurred in 4/6 and nausea in 2/6, but this is too small to establish dose response.
Phase 2 (220 randomized)	No serious adverse events reported. Related treatment-emergent events included headache, somnolence, nausea, abnormal dreams, and insomnia, generally low and sometimes also present on placebo. No active Stage-1 discontinuation was attributed to intolerance.	Short-term tolerability in selected MDD participants appears acceptable within the trial setting. It does not answer long-term safety, rare adverse events, interactions, or healthy use.
Unpublished follow-up	NCT02724735 planned a 6-month cohort with self-rated depression, cognition / physical functioning, sexual function, suicidality and discontinuation measures. Registry status is UNKNOWN and no results were posted or published.	These outcomes were planned, not evidence. Their absence leaves key experiential and longer-term questions unanswered.

Appraisal and synthesis

The science is most compelling at the **phenotype** level. NSI-189 repeatedly changes plasticity-associated endpoints across injury and metabolic disease models, with oral delivery in several studies. The most reproducible themes are neural progenitor / structural markers, synaptic plasticity, and functional behavior. The mitochondrial findings in diabetic nerve and cortex add a separate but still associative pathway.

The weak point is translation. There is no confirmed molecular initiating target, no verified human brain exposure or pharmacodynamic biomarker, no human hippocampal-volume effect, and no independent clinical replication. The larger MDD study makes the key corrective point: a compound can show plausible biology and positive secondary outcomes yet fail its main clinical endpoint. The Phase 2 paper was published by investigators with ties to the developer; this does not invalidate it, but it increases the importance of independent replication.

What is promising: oral activity in several animal models; durable post-treatment behavioral signal in the irradiated-rat experiment; explicit pro-plasticity data in Angelman-model tissue and mice; a low-dose, self-report / selected cognition signal in MDD. **What prevents a stronger conclusion:** target ambiguity, absence of exposure matching, diverse disease models, sponsor involvement, primary clinical null result, and no published long-term / healthy-user experience data.

What we do not know

- Whether NSI-189 improves any cognitive domain in healthy people, or whether effects in MDD are separable from depression symptom change.
- Its initiating molecular target, receptor occupancy, human brain distribution, or the concentration required for the laboratory effects.
- Whether the 40 mg signal is real, why 80 mg was not similarly positive, or whether an independently run trial reproduces any secondary cognitive outcome.

- Longer-term safety, withdrawal, dependence, mania / hypomania risk, sexual effects, fertility, cardiovascular effects, drug interactions, or rare harms.
- A credible qualitative account of what NSI-189 feels like. Online anecdotes are not included as evidence because no published, systematic study validates them.

Search strategy

Search date: 17 August 2026. Structured searches were performed in PubMed E-utilities, Europe PMC, OpenAlex, Crossref, ClinicalTrials.gov, and openFDA FAERS. Core queries: "NSI-189"[All Fields]; "NSI-189" AND (neurogenesis OR mechanism OR preclinical); "NSI-189" AND (qualitative OR interview OR "case report" OR "patient-reported" OR pharmacovigilance); and exact FAERS medicinal-product searches NSI-189 and NSI 189. The screen included primary human trials, primary animal / cellular experiments, registry records, and disconfirming evidence. Reviews were used only to locate primary studies and are not cited as evidence. No retraction affecting a key cited NSI-189 study was identified.

Sources

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U.S. Food and Drug Administration, openFDA Drug Adverse Event API. Exact medicinal-product searches for NSI-189 and NSI 189, checked 17 August 2026. No matches returned; this is not a safety finding. <https://open.fda.gov/apis/drug/event/>